



Particle fragmentation based on strain energy field

Yupeng Jiang¹ · Fernando Alonso-Marroquín¹ · Hans J Herrmann² · Peter Mora³Received: 28 February 2020 / Published online: 31 July 2020
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Abstract

We propose a method for the simulation of particle fragmentation based on the calculation of the strain energy field inside the particle. The topography of strain energy is calculated in terms of internal stress using the principles of damage and fracture mechanics. Numerical calculation of the energy field's ridges is used to determine the breakage criterion as well as the shape of the post-breakage fragments. This method provides a physical-based understanding of the breakage effect in granular material.

Keywords Particle breakage · Fragmenting model · Strain energy field · Granular material

1 Introduction

The mechanical properties of granular materials are fully governed by its discrete components and the contact forces. Breakage of a single particle, ranged from coarse sand (4–2 mm) to boulders (≈ 300 mm), could be caused by the strong compressive loading exerted by its surrounding environment. Breakage is affected by particle shape [1, 2], the number of particles [3–5], and the distribution of contact forces [1, 6]. In granular materials, particle breakage is commonly observed, which has a major effect on the mechanical properties, such as density [6], and compressibility [7]. As a collective response to particle breakage, the particle size distribution (PSD) [3–7] and normal compression line (NCL) [4–7] are adopted as indicators of the breakage effect. Their relations with the state variables have been studied through experiments [1–5, 7], which provides a solid foundation for understanding particle breakage. However, because of the difficulties of direct laboratory measurements, the relation between a single breakage event and the change of macro-scale mechanics is still poorly understood.

The particle replacement paradigm of the discrete element method (DEM), i.e. replacing the particle by its fragments when it reaches a breakage criterion, is used in many numerical models [8–14]. The existing methods possess high computational efficiency and the ability to reproduce PSD and NCL that agree well with experimental results [8]. Generally, two major attributes must be considered when developing a particle replacement model: a proper breakage criterion and a realistic approach for the determination of the geometry of post-breakage particles.

The existing breakage criteria are based on the averaged stress tensor. Particle breakage happens if the maximum principle stress or invariants of the stress tensor, or a linear combination of them, reaches a critical value. This paradigm has been adopted by many DEM-based breakage simulations due to high computational efficiency. The averaged particle stress tensor σ can be calculated as:

$$\sigma = \frac{1}{V} \sum_{n=1}^N (\mathbf{x}_n^c - \mathbf{x}_n^p) \mathbf{F}_n \quad (1)$$

The average stress is given in terms of the contact forces $\mathbf{F}_n$ that is exerted on the particle, the position of the centre of mass $\mathbf{x}_n^c$, the contact location $\mathbf{x}_n^p$, the total number of contacts N , here call *coordination number*, and V the volume of the particle. The stress tensor σ can be used for stress-based breakage criteria such as the von-Mises, the hydrostatic, or the maximum principal stress criterion. A typical use of Eq. 1 is to calculate the particle stress stemming from two diametrically opposite forces of magnitude F in the y -direction. For a spherical particle with a radius r

✉ Yupeng Jiang
yupeng.jiang@sydney.edu.au

¹ School of Civil Engineering, The University of Sydney, Sydney, NSW, Australia

² PMMH, ESPCI, 17 quai St. Bernard, 75005 Paris, France

³ College of Petroleum Engineering and Geosciences, King Fahd University of Petroleum and Minerals, Dhahran 31261, Saudi Arabia

the only non-zero stress component is $\sigma_y = (6F/r^2\pi)$. For a cylindrical particle with a radius r and thickness L , the non-zero stress component is $\sigma_y = (2F/r\pi L)$. The commonly used particle breaking criterion assumes that the particle breaks when the maximum principal component of the stress reaches a critical value σ_{cr} . To evaluate this critical particle strength the smaller is stronger principle is also applied. This is based on the experimental evidence [15–17] that the particle strength is higher when the particle is smaller $\sigma_{cr} \sim d^{-b}$, where b is a positive exponent [17–19].

Despite its computational efficiency, the averaged stress tensor oversimplifies the stress distribution inside the particle, which is also known as intra-granular stress [20] or sub-particle stress [21]. Overlooking the internal stress distribution causes a large deviation from the actual strength values obtained from finite element analysis and laboratory measurement [11, 12]. The averaged stress tensor also offers little information for determining the geometry of post-breakage particles. A better resolution of particle breakage can be obtained by evaluating the stress field inside the particle, here called sub-particle stress. The need for higher resolution breakage models is justified for the Brazilian test, where a cylindrical rock specimen is subjected by two opposite forces. The analysis of the sub-particle stress predicts a tensile failure at the centre of the particle given by $\sigma_t = (F/r\pi L)$, where σ_t is the maximal tensile stress of the specimen. This result is similar to the stress calculated by the averaged stress method in the paragraph above, except for a factor of 2. However, the stress average technique predicts a compressive failure, while the Brazilian test shows tensile failure. These deficiencies motivate us to propose an advanced particle breakage criterion based on the sub-particle stress.

In this paper, we develop a particle breakage method that is fully governed by the sub-particle stress state. An advanced physical perspective for the particle breakage is first introduced based on the principles of Griffith fracture mechanics [22]. The relationship between damage initiation, the orientation of fracture planes, and the sub-particle stress is discussed. The breakage criterion, as well as the fragmenting method, are then defined using the strain strain energy field function proposed by Lemaitre and Chaboche [23, 24]. Results are rigorously analysed with the experimental data of existing studies. The main purpose of this study is to introduce a new perspective of particle breakage based on sub-particle stress.

The paper is organized as follows: the theoretical background and breakage model are presented in Sect. 2. The validation for the single-grain prediction of particle breakage test is provided in Sect. 3. The discussion and general conclusions are made in Sect. 4.

2 Breakage model

2.1 Damage-fracture model

In our model, Griffith's elastic fracture mechanics [22] is regarded as the basic model for crack propagation. The particle is assumed as a *continuum material* (Fig. 1) that contains evenly distributed micro-scale elliptical cracks. The orientation and length of these flaws are random. The size and density of cracks are low enough so that the continuum mechanics approach is still applicable at the macroscale. Thus, the stress and strain field can still be calculated based on linear elasticity.

A distribution of strain energy density, here called strain energy field, is generated within the particle when external loads are applied. With the increase of strain energy, some local cracks are activated and start growing and coalescing with each other. Eventually, a group of activated cracks develops into a macro-scale fracture and causes splitting of the particle. This process describes the whole process of the single-particle breakage from the initial loading to the final breakage. The breakage criterion, the shape of replacement fragment, and the calculation of particle strength of new fragments will be defined based on this perspective.

In comparison to the existing theories, this physical perspective of breakage has several improvements. The breakage of the particle is regarded as a collective effect of crack growth [25] caused by the increase of strain energy. The size effect of particle strength, as an experimentally validated phenomenon, can still be incorporated into the theory since the maximum size of the crack is proportional to the particle size. The assumption of continuum material allows the efficient calculation of the stress field by linear elasticity. In this spirit, we do not consider the detailed properties of the micro-cracks and the dynamic progress of their propagation. In reality, the

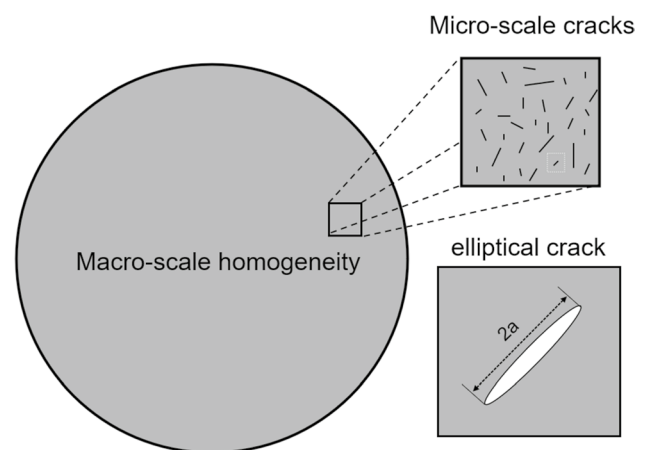


Fig. 1 The material model for the particle at different scales

mechanical properties of the particle certainly change with the growth of the cracks. However, this progress occurs in a short time and over a small scale compared with the quasi-static deformation of the granular system.

2.2 Breakage criterion

Under a certain loading condition, the activation of a crack is fully determined by its size and orientation angle. The critical stress of the activation decreases with the increase of the half-side of the crack a as shown in Fig. 1 and varies with the orientation angle. For an equal crack size, the configuration that produces the largest stress at the tip of the crack defines the *critical angle* [26]. If the local strain energy is sufficiently high, the largest local crack in the critical angle will be the first to grow or be activated. This crack is defined as critical crack in this paper.

The strain energy field function $\psi(\sigma, \nu)$ depends on the magnitude of the local elastic energy, where $\sigma = \sigma(x, y)$ is the Cauchy stress tensor and ν is the Poisson ratio. Similar to the von-Mises stress, the strain energy field function is a scalar value of the stress-based indicator for the elastic energy and independent of the material modulus. Now let us define a lower bound of ψ :

$$\epsilon \leq \psi(\sigma, \nu) \quad (2)$$

where ϵ is a material parameter that serves as the lower limit for the activation of a local critical crack. The sketch for the strain energy field of a two-dimensional disc under a diametric loading is shown in Fig. 2. The purpose of ψ is

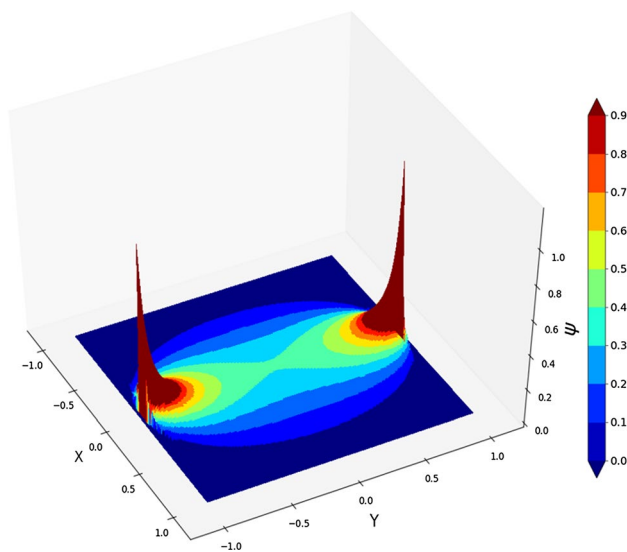


Fig. 2 The strain energy field of an elastic disc under diametric loading in the three-dimensional space consists of two spatial dimensions x , y , and the value of ψ

to provide a measurement for the elastic energy distribution over the two-dimensional space.

The value of ϵ represents the threshold needed to generate damage in the particle; no local crack will start to grow if the value of ψ is smaller than ϵ . Any area inside the particle that does not satisfy Eq. 2 can be regarded as an *intact domain*. If the local energy equals or exceeds this limit the critical cracks are activated. The domain of the particle in this state is defined as the *damaged domain*. The contour line $\psi(\sigma, \nu) = \epsilon$ is the boundary between the damaged and intact domains on the particle. The value of ϵ governs the characteristic strength of the particle; if $\epsilon = 0$, the particle is damaged under any non-zero contact forces; if $\epsilon = +\infty$, no damaged domain can be generated and the particle is unbreakable. Here, ϵ is calibrated with the maximum tensile stress at the centre of the disc under diametrical loading.

The evolution of the strain energy field under increasing external load and the growth of the damage zones is illustrated in Fig. 3. Here we plot the strain energy field along the line connecting the two load points in the Brazilian test. With an increase of the external loads, the damaged domains will expand and eventually percolate through the intact domain. The red dots $\psi = \epsilon$ are the boundary between the intact segment and the damaged segments in Fig. 3b–d. We assume that the breakage occurs if the damaged domains cover the entire central line (Fig. 3d). For the

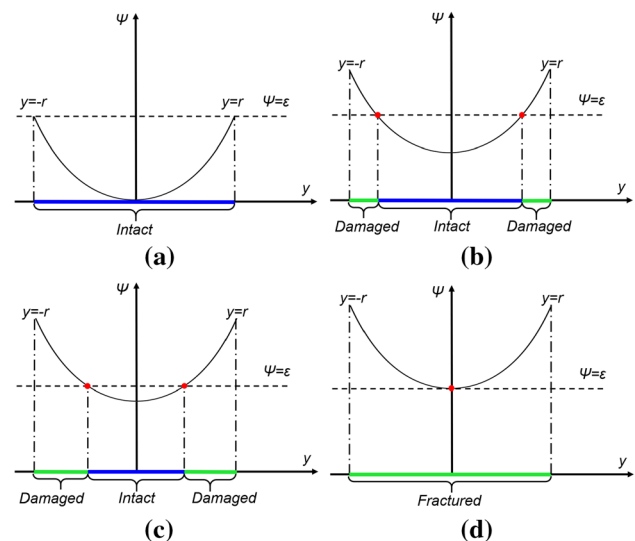


Fig. 3 The variation of the strain energy field and its corresponding material state in one dimension. The value of ψ is calculated along the central line ($x = 0$) of the two-dimensional Brazilian test with a radius of r . **a** no part of ψ is above ϵ , the whole domain is intact. **b** the strain energy field increases and the domains with $\epsilon \leq \psi$ are in the damage state; the red dots indicate the boundary between the damaged and intact state. **c** The value of ψ further increases; the damaged domains expand and the intact domain shrinks. **d** The damaged domains cover the entire central line and breakage occurs (color figure online)

two-dimensional calculation, the red dots become a contour line that separates the damaged and intact domains. The particle breaks if the damaged domain contains a line that geometrically separates the particle. This line is regarded as the fracture plane. In this way, the breakage criterion delivers the geometry of the new fragments through the stress field.

The optimal choice of ψ is still an open question. Its form needs to be associated with the elastic strain energy. Here we derive the form of ψ purely based on the elastic strain energy density W_e :

$$W_e = \frac{1}{2} \boldsymbol{\sigma} : \mathbf{e} \quad (3)$$

where $\boldsymbol{\sigma}$ and $\mathbf{e}$ represent the stress and strain tensor respectively, the symbol ‘:’ is the double dot product operator in tensor analysis. Assuming linear elasticity, σ_i and e_i in the principal direction follow the relationships:

$$e_i = \frac{1}{E} [\sigma_i - (\sigma_j + \sigma_k)] \quad (4)$$

where E is Young’s modulus and ν is Poisson’s ratio. The principal strain $\mathbf{e}$ in Eq. 3 can be replaced by Eq. 4 to obtain:

$$W_e = \frac{1-2\nu}{6E} \underbrace{(\sigma_1 + \sigma_2 + \sigma_3)^2}_{9\sigma_h^2} + \frac{1+\nu}{6E} \underbrace{[(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2]}_{2\sigma_v^2} \quad (5)$$

where σ_h and σ_v are the hydrostatic stress and the von-Mises stress. The expression for W_e is simplified as:

$$W_e := \frac{1}{2E} \left[\frac{2}{3} (1+\nu) \sigma_v^2 + 3(1-2\nu) \sigma_m^2 \right] = \frac{\psi^2}{2E} \quad (6)$$

where ψ can be seen as the equivalent stress associated with the elastic strain energy of the volume element. Therefore, this form of ψ is taken as the indicator of the elastic strain energy:

$$\psi(\boldsymbol{\sigma}, \nu) = \sigma_v \left[\frac{2}{3} (1+\nu) + 3(1-2\nu) \left(\frac{\sigma_m}{\sigma_v} \right)^2 \right]^{1/2} \quad (7)$$

The value of ψ can be conveniently calculated from the Cauchy stress tensor and considers both the influence of hydrostatic stress for brittle damage and of the von-Mises stress for plastic behaviour. Therefore, it is adopted as the elastic energy indicator for the activation of a local critical crack. The derivations of ψ correspond to the work of Lemaitre and Chaboche [23, 24]. They obtained Eq. 5 based on the thermodynamic potential of continuum mechanics and adopted it as equivalent stress to determine the elastic energy release rate associated with the local rupture or crack

initiation in damage mechanics. Conceptually this quantity shares certain similarities with our physical perspective as an energy threshold for crack growth.

The calculation of stress field is relatively easy. It can be regarded as a boundary problem with a pure Neumann (traction) boundary condition. Equation 8 states that the stress field inside the particle can be represented by a linear combination of fundamental solutions, the boundary traction $\mathbf{t}$ and the displacement $\mathbf{u}$ [21]:

$$\sigma_{ij}(\mathbf{x}) = \int_{\Gamma} D_{kij}(\mathbf{x}) t_k d\Gamma - \int_{\Gamma} S_{kij}(\mathbf{x}) u_k, (i, j, k = 1, 2) \quad (8)$$

where the fundamental solutions D_{kij} and S_{kij} are given as:

$$D_{kij} = \frac{1}{r} \{ (1-2\nu) [\delta_{ki} r_{,j} + \delta_{kj} r_{,i} - \delta_{ij} r_{,k}] + 2r_{,i} r_{,j} r_{,k} \} \frac{1}{4\pi(1-\nu)} \quad (9)$$

$$S_{kij} = \frac{2}{r^2} \left\{ 2 \frac{\partial r}{\partial n} [(1-2\nu) \delta_{ij} r_{,k} + \nu(\delta_{ik} r_{,j} - \delta_{jk} r_{,i}) - 4r_{,i} r_{,j} r_{,k}] + 2\nu(n_i r_{,j} r_{,k} + n_j r_{,i} r_{,k}) + (1-2\nu)(2n_k r_{,i} r_{,j} + n_j \delta_{ik} + n_i \delta_{jk}) - (1-4\nu)n_k \delta_{ij} \right\} \frac{1}{4\pi(1-\nu)}, \quad (10)$$

where r is the Euclidean distance between the boundary point and the inner point $\mathbf{x}$; δ denotes the Kronecker delta function; $\mathbf{n}$ is the outward normal vector at the boundary point; and $r_{,i}$ denotes the calculation the first derivative of the corresponding variable i . The specific values of $\mathbf{t}$ can be obtained by dividing the contact forces with the size of their corresponding contact areas on the boundary, it serves as the initial boundary condition (Neumann condition). The displacement vector $\mathbf{u}$ of the whole boundary of the particle is unknown. It can be numerically obtained using the boundary element method [21]. After the displacement $\mathbf{u}$ is found, the Cauchy stress at any inner point $\mathbf{x}$ of the particle can be calculated by substituting $\mathbf{t}$ and $\mathbf{u}$ into Eq. 8.

Equations 9 and 10 are proportional to $1/r$ and $1/r^2$ respectively which indicates that the maximum deformation occurs at the point where the traction (source) is applied and decreases with the Euclidean distance from the source point. The strain energy particularly concentrates near the loading point and decreases quadratically with the Euclidean distance. It suggests that the damage domain should start from each source point and expand with an increase in the external loading. The fracture plane can be defined once damaged domains that originate from the source points connect with each other and particle breakage occurs.

The breakage criterion is shown in Fig. 4 based on Eq. 7; the damaged domain is illustrated in green. Similar to Fig. 3,

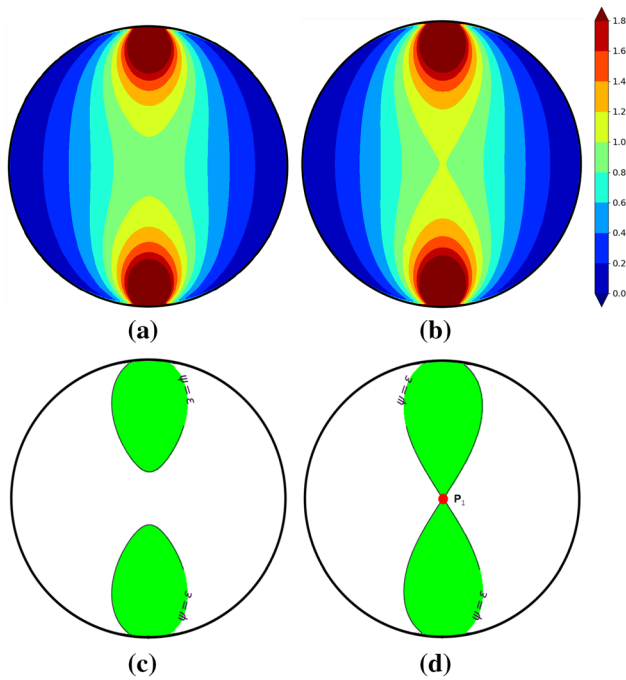


Fig. 4 The growth and coalescence of the damaged domains with the diametrical loading in two dimensions. The red dot represents the connection point and the colour denotes ψ as calculated from Eq. 3. The contour lines in **a** and **b** are the function ψ normalized by ϵ . **a** Shows the strain energy field of an insufficient loading; **b** the strain energy field of the breakage loading; **c** the boundary of ϵ in **a** is not connected, (no breakage); **d** the boundary of ϵ in **b** is connected due the increase of the contact forces so that breakage occurs (color figure online)

the damaged domains are generated inside the particle and bounded by the intact area $\psi < \epsilon$ in Fig. 4c since the force is not large enough. With the increase of the forces, the damaged domains expand and a line inside the particle can be defined when damaged domains that originated from opposite source points connect at the centre of the particle in Fig. 4d. Such a path that has no intact area to block the growth and coalescence of the locally activated cracks inside the damaged domain; a macro-scale fracture will be formed. Based on these features, the breakage criterion for the diametrical loading is further defined by the existence of a *connection point* $\mathbf{P}_i$ that satisfies:

$$\begin{cases} \psi(\boldsymbol{\sigma}, \nu)|_{\mathbf{P}_i} = \epsilon \\ \|\nabla \psi|_{\mathbf{P}_i}\| = 0 \end{cases} \quad (11)$$

that is, the particle breakage occurs when the value of ψ at the connection point $\mathbf{P}_i$ equals ϵ , and the location of the connection point is a stationary point of the function ψ .

The determination of the connection points becomes complicated if the number of load points, here defined as coordination number (N), is larger than two. It can be

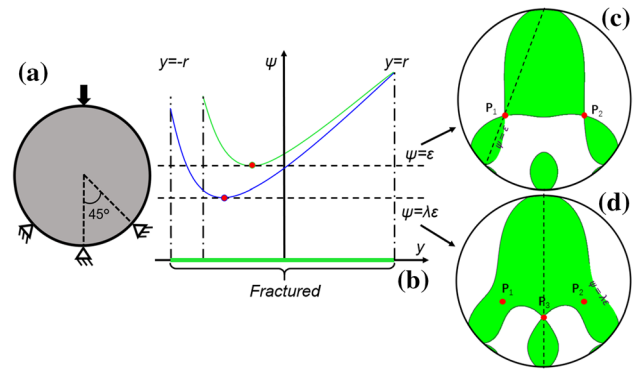


Fig. 5 The growth and coalescence of the damaged domains with a high coordination number ($N = 4$). **a** Load configuration. The black arrow represents the compressive force and the triangles denote the fixed boundary; **b** projection of strain energy fields. The orange and blue lines represent the y -axis projected strain energy field along the dash lines shown in **c** and **d**; **c** a damaged zone containing two connection points $\mathbf{P}_1$ and $\mathbf{P}_2$ resulting from $\psi = \epsilon$; **d** damaged zone containing all three connection points when $\psi = \lambda\epsilon$ (color figure online)

observed from Fig. 5 that the connection points increase as the N increases. The values of ψ at each connection points do not reach the threshold $\psi = \epsilon$ simultaneously. This assumption is justified by the experiments of Salami et al. [27]. Their experiment suggests that the particle breakage is a multi-stage process; the fracture planes are actually produced in a sequence before the equilibrium of the contact forces is broken by the generation of new fragments. In the particular case of the configuration shown in Fig. 5, the experiments show that the two breakage surfaces passing to the connection points $\mathbf{P}_1$ and $\mathbf{P}_2$ are created first. Then a secondary breakage surface passing to the point connection point $\mathbf{P}_3$ is produced due to the damage of the primary fragments. Other experiments [2, 28] confirm this progressive damage pattern. Cracks and voids in the damaged area expand and therefore cause the deterioration of the material prior to the final breakage.

To account for this multi-stage fragmentation process, we assume that the particle reaches first the breakage criteria $\psi = \epsilon$. Then the connection points $\mathbf{P}_1$ and $\mathbf{P}_2$ shown in Fig. 5c are calculated based on Eq. 7. These two points lead to two primary breakage surfaces. Then, it is assumed that the strength of the fragments is reduced by $\lambda\epsilon$, where $0 < \lambda < 1$. Next, the damage domain is recalculated using $\psi = \lambda\epsilon$, as shown in Fig. 5d. If a small enough value of λ is chosen, a third connection point $\mathbf{P}_3$ satisfying $\psi(\mathbf{P}_3) = \lambda\epsilon$ is obtained. A lower value of λ means a lower energy threshold, which may include more qualified connection points and produces more fractures. In Fig. 5 we use $\lambda = 0.91$ to obtain the three connection points that ultimately define the three fracture surfaces observed in the experiments.

2.3 Fracture planes

Here we propose a method to trace the fracture plane based on the damage-breakage perspective and the strain energy field ψ . This method is governed by the idea that the fracture planes have the tendency to cross the area with the highest local energy. Based on this criterion, we calculate the ridges of the strain energy field using the Hessian matrix, which consists of the second derivatives of ψ within the Cartesian coordinate system. The general direction of the fracture planes should correspond to the ridges of the energy field. This is underlined by the principle that the propagation and coalescence of micro-cracks have the tendency to stay within the area of high deformation energy. Therefore, the ridges of the strain energy strain are adopted here as a first-order prediction of the fracture planes [29]. A ridge point is identified as a point where ψ satisfies:

$$\frac{\partial \psi}{\partial u} = 0, \quad \frac{\partial^2 \psi}{\partial u^2} = 0, \quad \psi \geq \epsilon. \quad (12)$$

where u is defined as the direction of the eigenvector associated with the largest absolute eigenvalue of the Hessian matrix.

2.4 Discussion

In this section, we provide a discussion about the extent of our framework. In particular, we warn the reader on the limitations of the method to capture the effects of the hydrostatic pressure, and the lack of the history variables to account for damage and elastoplastic deformation.

First, let us note that our breakage model introduces the concept of the strain energy field that is fundamentally different from the classical concept of the failure surface. The inequation $\epsilon \leq \psi$ defines the general vicinity for the crack coalesce in the material, which confined by the energy threshold ϵ . The crack growth can only happen when the local strain energy represented by ψ fulfills this criterion. The strain energy field function does not equal to a determination of a failure on the material but an energy-wise qualification for crack coalesce. Therefore, different from the failure model, our method considers the spatial distribution of ψ to determine the breakage and the configuration of fracture planes.

Since particles in granular materials are mostly subjected to the point-like compression [2, 3], the crack growth under compressive loading needs further clarification. Under compressive stress, the stress field is locally modified by the crack generating tensile stress near the crack tips [26, 30]. The tensile stress depends on the length and orientation of the crack [26]. The maximum tensile stress σ_{max} occurs at the

crack tip and it is given by $\zeta_0 \sigma_{max} = -(\sigma_1 - \sigma_3)^2 / (\sigma_1 + \sigma_3)$, where $\zeta = \zeta_0$ is the elliptical coordinate of the crack width; σ_1 and σ_3 represent the maximum and the minimum principal stress components [26]. Thus, a crack can still activate and grow under compression, except under pure hydrostatic load ($\sigma_1 = \sigma_3$).

We assume a homogenous distribution of cracks with uniform size and orientation distribution. Thus the crack activation has no anisotropy and cracks coalesce, as a collective effect of activation, can be regarded as solely dependent on the elastic strain energy. Function ψ , therefore, can properly estimate the damaged area without considering the detailed tensile/compressive state on individual cracks [31]. It also needs to be pointed out that the anisotropy of the material strength is deemed to be less important for the fragmentation criterion and fracture planes by the experimental studies [32]; the direction of the fracture planes is dominated by external contacts and stress state [33].

The formulation does not consider the influence of material properties such as the cohesion and angle of friction. For this reason, the model does not distinguish between compressive and tensile strength. Figure 6 shows that the envelope of strain energy field $\psi = \epsilon$ in the stress space. This is an axially symmetrical, smooth closed surface with the original point as the centre. The projection of this surface in the plane is also shown in Fig. 6a. This projection implies that compressive and tensile strength are the same in our model. This is different from real materials, where the compressive strength for rocks is often distinctively high than the tensile strength. This limitation can be overcome by considering the plastic component of the strain, which is neglected in this formulation. For example, the Drucker-Prager elastoplasticity model can be included in the stress-strain relation. This will allow us to include the cohesion and angle of friction into the elastic energy formulation to fully account for the material strength parameters.

Our model also does not include the variable D in continuum damage mechanics [23, 24]. This variable may need

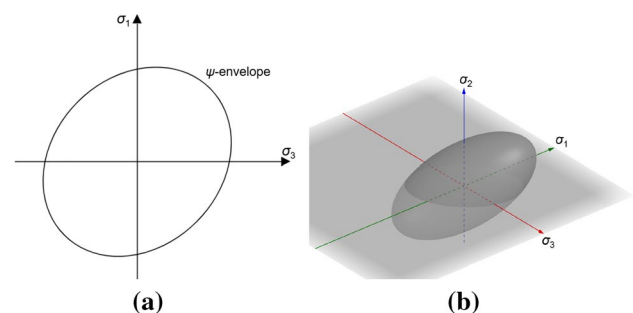


Fig. 6 Normalized two and three-dimensional envelopes of strain energy field function $\psi = \epsilon$ ($\nu = 0.2$) **a** 2D projection, **b** 3D envelopes

to be considered for the long-term deformation of the granular material, where particle breakage does not be instantly caused by contact forces but deteriorating over time. The severity of the breakage might be underestimated without considering the evolution of D on the individual particle. While we ignore the factor D for the single-particle test, it can be incorporated into the numerical implementation in the granular material such as the discrete element method (DEM). The value of D can be approximated at each certain amount of time step with $D = V_d/V$, where V_d and V are the damage volume and total volume of the particle respectively. If the particle is not fragmented at the given time step, its damage can be carried by the parameter D to account for future breakage.

It also needs to be noticed the calculation of the ridges of the strain energy field by Eqs. 11 and 12 not longer work when pure hydrostatic loading is applied to the particle. In this case ψ becomes constant over the entire particle. Then the first derivative is zero and the ridge of the strain energy field disappears. When modelling granular materials, a pure hydrostatic loading is quite an extreme loading condition due to the presence of the force chains [2–5]. Our method is suitable for dry or unsaturated granular material (dry sand, coarse grain, and railroad ballast), but should be modified for the saturated granular material (e.g. pebble layer at the bottom of rivers) when calibrating the value of ϵ since the pore pressure often prevents the breakage yet increase the elastic strain energy.

3 Validation of single particle breakage

The breakage scheme is validated with different values of coordination number on a two-dimensional Brazilian disc. The basic configuration of the six tests used in their paper [27] is shown in the first column of Fig. 7. The experiment used a disc of cement stone as a sample in the Brazilian test. The experiment used discs made out of the mix of cement and Hostun siliceous sand as the sample in the tests. The size of the samples in the experiment is 52 mm in diameter and 21 mm in width. The active compressive load, which is illustrated through the black arrow, is applied at the top of the samples. The rest of the contact points are represented by linear clamping bars, which passively generate reaction forces under compression. The boundary element method is used to calculate the stress field. In the numerical simulation, the active and passive contacts are regarded as boundary compression and zero-displacement boundary condition respectively. The energy threshold ϵ is calibrated using the value obtained from the diametrical test.

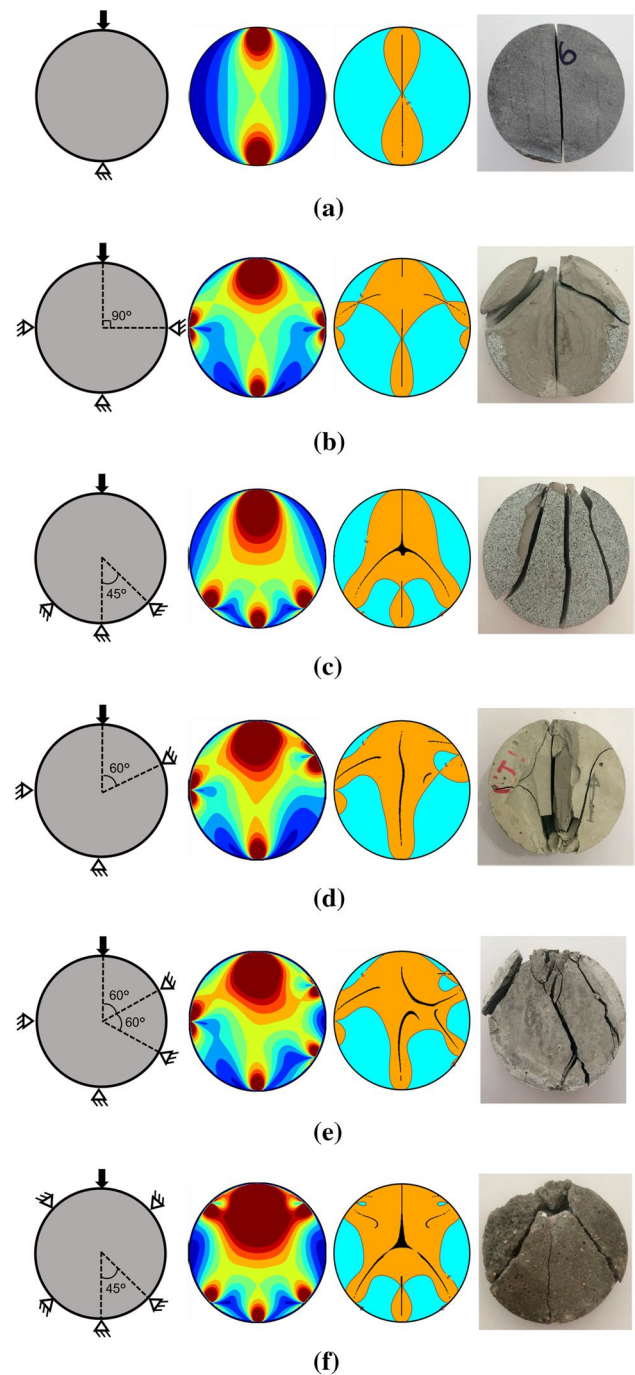


Fig. 7 Comparison between the numerical and experimental results for the fractures of a circular particle (a–f); the first row shows the configuration of each test with different coordination numbers. The black arrow denotes the compressive force while the tailed triangle is the place where a zero-displacement boundary is applied; the second row contains the contour lines of $\psi = \epsilon$ generalized by ϵ ; the third row shows the numerical fracture planes provided by our method; the last row exhibits the images of the fragments from the experiment

Results of critical force and fragment shape are rigorously compared and analysed with the existing experimental data and images produced by Salami et al. [27]. The results demonstrate that our sub-particle breakage analysis could provide valuable information about both breakage conditions and the shape of fragments.

3.1 Validation for the shape of fragments

The approximate shape of the fragment can also be retrieved from the numerical analysis of the strain energy field calculated from the particle stress field. The third row of Fig. 7 shows the damaged domains (orange colour) at the moment of breakage and the separation lines; the experimental images of the fractures are given in the last row of Fig. 7 for a direct comparison. It can be observed that the fracture planes obtained from the sub-particle breakage scheme agree well with the experimental images for each coordination number. Both the orientation and the number of fractures are properly captured. Especially, the pattern of fracture planes connecting different contact points is successfully predicted. It must be pointed out that the connection between a pair of contact points is a dominating feature for solid particle breakage, where the point-loading condition is common. This phenomenon has been confirmed by the experimental studies of Artoni et al. [32] on a single-particle test. The result further validates our strain energy field model since it provides a proper explanation for the geometrical pattern of the fracture planes.

The agreement of fracture planes is particularly good for the cases of (a), (b), (c) and (f). Minor differences are found in cases (d) and (e), where the numerical scheme provides fewer fragmentations than the experiment and have different shapes. Such differences are mainly due to the fact that our model does not consider the variations of the strain energy field that happen during the fracturing process. In reality, the fractures interact and will affect the generation and propagation of each other through the changes in the strain energy field. This process means that one fracture is able to attract, repel or arrest other fractures in its vicinity. Fracture propagation can in principle be captured by dynamically simulating the entire process, which is computationally expensive. In our method, the particle is treated as an intact entity before its final breakage and fragmentation. It approximates a strongly nonlinear process by a static strain energy field. The differences are acceptable considering the improvement in computational efficiency and the reduction of complexity brought by this method.

Other factors such as material heterogeneity [33, 34] and the relative displacement between the sample and constraint points could also contribute to the disagreement between numerical and experimental results. Hence, minor differences are expected between the actual breakage and

numerical simulation. The results indicate that such differences can be properly controlled to reach an optimized compromise of accuracy and computational efficiency. More importantly, the results further validate the assumption that the strain energy field is an important player of particle breakage.

3.2 Validation of the breakage criterion

The final breakage load for each case is defined as the critical force. The comparison of the critical forces at different coordination number is presented in Fig. 8. The analytical values in Salami et al.'s [27] study are calculated based on a formula proposed by Sukumaran et al. [35], which includes an empirical nonlinear factor to approximate the effect of the coordination number on the critical force.

It can be observed from Fig. 8 that each group of the critical force obtained from experimental, analytical and numerical tests increases with N . Such a result shows that the particle shows a stronger breakage resistance when it has more loading contacts. This is consistent with the conclusions made by several other studies about the effect of the coordination number. A rigorous analysis of this phenomenon is provided by Jiang et al. [21] via a series of two-dimensional numerical studies of the particle stress field, which concluded that the increase of the coordination number can decrease the maximum principal stress and strain energy inside the particle. However, this study is only valid for a perfectly circular particle since the maximum principal

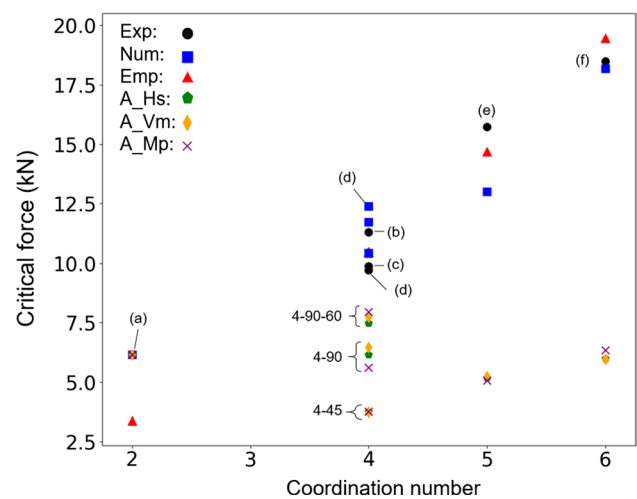


Fig. 8 Critical force for different coordination numbers. The experimental results (Exp) are represented with black dots; the empirical results (Emp) are denoted with the red triangle and our numerical results (Num) are represented with blue squares. The results calculated from the averaged stress tensor are also provided for comparison, which includes hydrostatic stress (A-Hs), von-Mises stress (A-Vm) and maximum principal stress (A-Mp) (color figure online)

stress increases with the coordination number for elliptical particles.

The results provided by our method show a rather good agreement with the experimental data. Our method is calibrated with the experimental data $N = 2$ and particularly useful for $N = 4, 6$, where the difference between the empirical formula and experiment data is distinctively large. The largest deviation of our method from the experimental data is 15% for $N = 5$. The critical force calculated from the averaged stress fails to provide a proper approximation for all the cases of $2 < N$. It cannot reflect the effect produced by the increase of the coordination number, which further proves the errors generated by using the averaged stress tensor are not negligible even for a circular particle.

Our results prove that our basic assumptions are valid for various loading situations. Both the empirical formula and our numerical method can be applied to approximate the breakage of circular particles under various coordination numbers. However, for the case of irregular particles, the validity of the empirical formula becomes questionable. The geometrical properties, such as void ratio and angularity, have a strong influence on the breakage resistance. It is difficult to generalize all the situations with one single empirical formula. Our numerical method, which is based on the sub-particle stress field analysis and the deformation strain energy field, seems more suitable to provide the criteria for the breakage of irregularly shaped particles.

4 Conclusions

In this paper, we proposed a new physical perspective for particle fragmentation. This method helps to solve two of the existing limitations of the use of the averaged stress tensor: the oversimplification of particle strength and the lack of sub-particle stress field analysis. Our framework provides an alternative way with an optimized compromise between the computational efficiency and accuracy for calculating the particle breakage at both individual particle and granular level. The strain energy field is derived from the theory of damage mechanics. The particle breakage is interpreted as the result of the percolation of the damage domains generated by the strain energy. The strain energy field of the particle under external loading is given by the sub-particle stress field. The distribution of the damaged domain on the particle determines the breakage criterion and the geometry of the fracture planes.

The breakage limit and the geometry of the fracture planes are validated with the experimental results of the Brazilian test under various coordination numbers. The comparison indicates that both the critical force, calculated from the breakage criterion, and the shape of the fragments, agree well with the experimental data. In conclusion, our particle

breakage theory unifies the breakage criterion, shape of the new fragments, and the sub-particle stress field under one theoretical frame. It avoids the use of empirical parameters and the problems caused by oversimplification with averaged stress tensor. Therefore, the framework is expected to make a good contribution to the calculation and numerical simulation of particle breakage.

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Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

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